

Z-CURVE 2.0**Tier 2a :: p-Hacking / Selective Reporting**

At a glance: z-curve 2.0 converts all reported test statistics in a literature to z-scores, then fits a finite mixture model to the full distribution — both significant and non-significant results. From this fit it estimates two distinct quantities unavailable from any other method: the expected replication rate (ERR), which predicts how often significant findings would replicate in direct replications, and the expected discovery rate (EDR), which estimates the proportion of tested hypotheses that are actually true. Requires test statistics and degrees of freedom; no raw data needed.

1. Overview

z-curve 2.0, developed by Frantisek Bartos and Ulrich Schimmack and published in 2022, addresses a fundamental limitation shared by p-curve, p-uniform, and the caliper test: all three methods work exclusively with the significant portion of a literature's p-values, discarding the information contained in non-significant results. This is a substantial loss. Non-significant results, even when unpublished, leave traces — in registered reports, in preregistered studies that report all outcomes, in clinical trial registries, in meta-analytic databases that include non-significant effects. And when they are available, they are highly informative about the statistical properties of the research program that generated the published significant results.

z-curve takes a different approach. Rather than conditioning on significance, it models the full distribution of test statistics — significant and non-significant — as a mixture of noncentral distributions. Each component of the mixture corresponds to a different level of statistical power. By fitting this mixture model to observed z-scores, z-curve simultaneously estimates what the power distribution of the studies looks like and what that implies about the replication prospects of the significant findings and the true discovery rate of the research program.

The two key outputs — ERR and EDR — answer questions that are at the center of the reproducibility debate but that no prior tool could address quantitatively. The ERR answers: if we ran direct replications of all the significant findings in this literature, what fraction would replicate at $p < .05$? The EDR answers: of all the hypotheses being tested in this research program (including the ones that produced non-significant results that were never published), what fraction are actually true? The gap between ERR and EDR — and the gap between EDR and the observed discovery rate in published papers — quantifies the severity of publication bias in the literature.

2. Methodological Overview**Step 1 — Convert all test statistics to z-scores**

Every test statistic (t, F, chi-squared, r) reported in the studies under analysis is converted to an equivalent two-tailed z-score using the inverse normal transformation. For a t-statistic with df degrees of freedom, the conversion is: $z = \text{qnorm}(1 - \text{pt}(|t|, \text{df})/2)$, where pt is the t-distribution CDF. This places all results on a common scale and allows the mixture model to be fit in z-space regardless of the original test type.

The resulting set of z-scores spans from 0 to some upper bound (typically capped at $z = 6$ to exclude extreme outliers that could destabilize the fit). The critical region for the standard alpha = .05 threshold corresponds to $|z| > 1.96$. Results above this threshold are significant; those below are non-significant.

Step 2 — Fit the finite mixture model

The observed z-score distribution is modeled as a mixture of K noncentral half-normal distributions, each representing a cohort of studies conducted with a specific level of statistical power. The mixture has the form:

$$f(z) = \sum^K \pi^k \cdot g(z \mid \lambda^k)$$

Where K is the number of mixture components, π^k are the mixing weights (summing to 1), and $g(z \mid \lambda^k)$ is a noncentral half-normal density with noncentrality parameter λ^k . Each λ^k corresponds to a power level.

$$\text{Power}(\lambda^k) = P(|Z| > 1.96 \mid \lambda^k) = 1 - \Phi(1.96 - \lambda^k) + \Phi(-1.96 - \lambda^k)$$

The power associated with each mixture component. Components with larger λ^k represent high-powered studies; smaller λ^k represent low-powered studies.

The mixture model is fit using maximum likelihood estimation (or EM algorithm) over the observed z-score distribution. The fitting accounts for selection: because non-significant results are less likely to be published, the likelihood is adjusted by the probability of selection under the estimated power distribution. This selection correction is what separates z-curve from naive mixture model fits and what allows it to estimate the full distribution of true power — not just the power of what was published.

Step 3 — Compute ERR and EDR

Once the mixture model is fit and the power distribution estimated, the two key quantities follow directly:

ERR = $\sum^k \pi^k \cdot \text{Power}(\lambda^k)$

Expected Replication Rate: the weighted average power across all mixture components. Answers: if a direct replication were run for each significant finding, what fraction would replicate at $p < .05$?

EDR = **ERR** / **ODR**

Expected Discovery Rate: ERR divided by the Observed Discovery Rate (the fraction of all tested hypotheses that produced significant results). Answers: what proportion of all tested hypotheses are actually true, including those with non-significant results that were never published?

Bias = **ODR** - **EDR**

Publication bias index: the gap between what was observed in the published record (ODR) and the estimated true discovery rate (EDR). A large positive value indicates severe publication bias.

The z-curve plot — a histogram of all z-scores with the fitted mixture model overlaid — provides a visual summary. The fitted curve shows the expected distribution under the estimated power mix. Deviations of the observed distribution from the fitted curve, particularly the spike near the significance threshold from p-hacking or the trough just above it from publication bias, are visually apparent.

3. Interpreting ERR and EDR

The relationship between ERR and EDR reveals the statistical health of a research program in a way that no single number can capture. Four canonical patterns are worth understanding:

High ERR, High EDR	A well-powered research program with genuine effects and low publication bias. Studies are adequately powered, most tested hypotheses are true, and the published record is approximately representative of all tested hypotheses. Target state for a healthy field.
High ERR, Low EDR	A research program where published effects are real and well-powered, but there is massive publication bias — many hypotheses are being tested and only the successful ones appear in print. The significant findings will replicate, but the published literature is a heavily filtered view of an extremely low base-rate testing environment.
Low ERR, High EDR	A research program where most tested hypotheses are true but studies are systematically underpowered. Published effects are real but fragile — they will often fail to replicate because the original studies were just barely significant. The solution is larger sample sizes, not better hypotheses.
Low ERR, Low EDR	The worst-case scenario: underpowered studies, low true positive rate, and likely substantial p-hacking to produce publishable results. This pattern characterized the replication crisis in social psychology — a low base rate of true effects combined with underpowered flexible analyses.

4. Example Application: Social Psychology and the Replication Crisis

Bartos and Schimmack (2022) applied z-curve 2.0 to a sample of test statistics from articles published in leading social psychology journals between 2010 and 2017. The z-curve plot showed a pronounced spike at the significance threshold — consistent with widespread p-hacking — and a rapid falloff of z-scores just above the threshold, consistent with publication suppression of non-significant results.

The fitted mixture model estimated an ERR of approximately 54% — meaning that if direct replications were run for all the significant findings in this sample, approximately half would replicate at $p < .05$. This aligns strikingly well with the empirical replication rate observed in the Open Science Collaboration's Reproducibility Project (2015), which found that approximately 36–50% of 100 psychology findings replicated. z-curve's estimate, derived entirely from the shape of the z-score distribution without running any replications, correctly predicted the empirical outcome.

The EDR estimate was substantially lower — approximately 31% — indicating that across all hypotheses being tested (including the non-significant ones that did not appear in print), only about a third were actually true. The gap between the ODR (the fraction of published results that were significant, approximately 95%) and the EDR (31%) quantifies the publication bias in the literature: a 64-percentage-point gap between what appeared in print and the estimated true discovery rate.

Subsequent z-curve analyses have been applied to clinical psychology (Fries & Frankenbach 2020), cognitive neuroscience, organizational behavior, management research, and economics. ERR estimates vary substantially across fields, from approximately 40% in some areas of social psychology to approximately 70–75% in cognitive psychology and experimental economics, consistent with known differences in study design and hypothesis testing practices across disciplines.

5. Requirements & Applicability

z-curve 2.0 is computationally and statistically more demanding than the simpler p-value distribution methods, but its requirements are manageable for any researcher with access to a bibliometric database and the R package zcurve.

Test statistics and df	The exact test statistic (t, F, z, r, chi-squared) and degrees of freedom for each reported hypothesis test. z-curve requires exact z-score conversion, which is only possible with the underlying statistic and df. Reported inequalities ($p < .05$) cannot be used and must be excluded.
Both significant and non-significant results	Unlike p-curve and p-uniform, z-curve benefits substantially from including non-significant results. The EDR estimate in particular is highly sensitive to the ratio of significant to non-significant results, which requires a sample that includes both. Literatures where non-significant results are systematically absent (severe file-drawer problem) will produce biased EDR estimates.
Large sample of test statistics	The mixture model fitting is unstable with fewer than approximately 100 z-scores. Reliable ERR and EDR estimates with narrow confidence intervals typically require 200 or more test statistics. z-curve is a field-level tool, not a study-level or small-literature tool.
Consistent test type	All test statistics should correspond to hypothesis tests of the same general type — directional tests of individual effects, not omnibus F-tests or goodness-of-fit statistics. Mixing qualitatively different test types distorts the z-score distribution and invalidates the mixture model.
Focal versus all tests	A critical analyst decision: using only focal hypothesis tests (one per study) versus all reported tests. Using all tests inflates the sample but includes exploratory and secondary analyses that may have different power characteristics than primary tests. The choice should be documented and sensitivity analyses conducted across both approaches.
Raw data required?	No. z-curve operates on reported test statistics. Automated extraction tools (statcheck, custom parsers) can build the z-score dataset from large literature samples. The zcurve R package handles all conversions and model fitting internally.

6. Limitations

- The EDR estimate is highly sensitive to the inclusion or exclusion of non-significant results. If non-significant results are systematically absent from the dataset (severe publication bias), the EDR will be overestimated — the model has no information about the null-result studies and must infer their frequency from the shape of the significant tail alone. Researchers should assess the completeness of their non-significant result coverage before interpreting EDR.

- The mixture model requires choosing K , the number of power components. z-curve 2.0 uses $K = 7$ components by default, which provides sufficient flexibility for most distributions without overfitting. However, the choice of K can affect estimates in edge cases. Sensitivity analyses across different K values are advisable when estimates are used for high-stakes conclusions.
- z-curve cannot diagnose the source of low power or low EDR. A low ERR could reflect underpowered studies, measurement error, researcher degrees of freedom, or genuine null effects mixed into a literature testing both true and false hypotheses. The tool diagnoses the statistical signature; interpreting what produced it requires domain knowledge.
- The tool is not appropriate for literatures with heterogeneous test types or mixed directional and non-directional hypotheses. Mixing omnibus tests, interaction terms, and simple main effects creates a z-score distribution that does not conform to the mixture model's assumptions.
- z-curve estimates come with substantial uncertainty. Confidence intervals on ERR and EDR are typically wide — 95% CIs of ± 10 –15 percentage points are common with samples of 200 z-scores. Point estimates should always be reported with their confidence intervals. Comparing ERR estimates across fields requires overlapping confidence intervals before concluding the fields differ.
- The model assumes that selection into the published record is determined solely by significance at $p < .05$. In practice, selection is more complex — effect size, sample size, novelty, and journal prestige also affect publication probability. Violations of the simple selection assumption bias the EDR estimate in ways that are difficult to assess.

7. How to Apply

Building the z-score dataset

Define inclusion criteria before extracting statistics. Use statcheck or a custom parser to extract test statistics from PDF files, or manually code statistics from a random sample of papers in the target literature. For each test, record the statistic type, value, and degrees of freedom. Convert all to z-scores using the zcurve R package's conversion utilities or the qnorm/pt functions directly.

Running z-curve in R

- Install and load the package: `install.packages('zcurve'); library(zcurve)`.
- Create the z-score vector: `z <- c(2.1, 3.4, 1.8, 2.9, ...)` or pass p-values directly via `z <- qnorm(1 - p/2)` for two-tailed tests.
- Fit the model: `fit <- zcurve(z)`. The default uses $K = 7$ components with the EM algorithm.
- Extract estimates: `summary(fit)` returns ERR, EDR, and their 95% bootstrap confidence intervals. `plot(fit)` produces the z-curve plot with the fitted mixture overlay.
- Run sensitivity analyses: `refit <- zcurve(z, method = 'density')` for the kernel density alternative, and compare ERR and EDR estimates across methods.

Reporting standards

Report the number of z-scores analyzed, the focal test selection rule, the ERR and EDR estimates with 95% confidence intervals, the observed discovery rate (ODR) for comparison, and the z-curve plot. State explicitly whether significant and non-significant results were both included, and if non-significant results were excluded, note that EDR estimates will be upwardly biased. Compare ERR to any available empirical replication rate data as a validation check.

8. z-curve 1.0 vs. z-curve 2.0

z-curve 1.0 (Schimmack & Brunner 2017) estimated only the ERR — it fitted the mixture model to significant z-scores only, producing an estimate of expected replication rate without an EDR. z-curve 2.0 extends the approach in two important ways: it incorporates non-significant z-scores into the likelihood, substantially improving the stability of the power mixture fit; and it derives the EDR from the fitted mixture and the observed discovery rate, adding a publication bias estimate that was entirely absent from the original version. In practice, z-curve 2.0 should be preferred in all applications — it is strictly more informative than 1.0 and the R package implements both for backward compatibility.

9. Related Tools

p-curve	Operates only on significant p-values; diagnoses evidential value and estimates average power without modeling the full z-score distribution. p-curve is more appropriate for small literature samples (20–50 studies) where z-curve's mixture model would be unstable. Run p-curve for evidential value, z-curve for field-level replication rate estimation.
p-uniform / p-uniform*	Produces a bias-corrected effect size estimate alongside an evidential value test; also operates only on significant results. Complementary to z-curve: use p-uniform for a bias-corrected effect size, z-curve for a bias-corrected replication rate and discovery rate.
Caliper test	Tests for a spike at the significance threshold; a simple diagnostic of threshold-seeking behavior. The caliper test's positive signal is consistent with the spike near $z = 1.96$ visible in z-curve plots of p-hacked literatures. The two tools are complementary diagnostics of the same underlying problem.
Excess Significance Test (TES)	Compares observed to expected significant result counts given study power. z-curve and TES both estimate expected power but use different methods: z-curve from the full z-score distribution, TES from assumed effect sizes and sample sizes. Convergence of z-curve's low ERR with a positive TES provides strong evidence of systematic reporting problems.
Open Science Collaboration (2015)	The empirical benchmark against which z-curve ERR estimates can be validated: 36–50% replication rate in psychology. z-curve's estimates for the same body of literature closely predicted this rate, providing the method's most important external validation.

Key Citations

- Bartos, F. & Schimmack, U. (2022). z-curve 2.0: Estimating replication rates and discovery rates. *Meta-Psychology*, 6. <https://doi.org/10.15626/MP.2021.2720>
- Schimmack, U. & Brunner, J. (2017). z-curve: A method for the estimating replicability based on test statistics in original studies (version 1.0). *OSF Preprints*. <https://osf.io/wr2ub/>
- Bartos, F. & Schimmack, U. (2020). zcurve: An R package for fitting z-curves. CRAN. <https://cran.r-project.org/package=zcurve>
- Open Science Collaboration. (2015). Estimating the reproducibility of psychological science. *Science*, 349(6251), aac4716. <https://doi.org/10.1126/science.aac4716>
- Simonsohn, U., Nelson, L. D., & Simmons, J. P. (2014). P-curve: A key to the file drawer. *Journal of Experimental Psychology: General*, 143(2), 534–547. <https://doi.org/10.1037/a0033242>
- Friese, M. & Frankenbach, J. (2020). p-hacking and publication bias interact to distort the discovery rate in psychological science. *PLOS ONE*, 15(5), e0233558. <https://doi.org/10.1371/journal.pone.0233558>
- Ioannidis, J. P. A. & Trikalinos, T. A. (2007). An exploratory test for an excess of significant findings. *Clinical Trials*, 4(3), 245–253. <https://doi.org/10.1177/1740774507079441>
- van Aert, R. C. M., Wicherts, J. M., & van Assen, M. A. L. M. (2019). Publication bias examined in meta-analyses from psychology and medicine: A meta-meta-analysis. *PLOS ONE*, 14(4), e0215052. <https://doi.org/10.1371/journal.pone.0215052>